QAOA · qBraid challenge · Qiskit Fall Fest Connecticut 2026

Solving Instant Insanity with constrained QAOA



86.7% of shots return the one correct tower on AWS SV1

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THE PROBLEM

Four cubes, one tower that works

- Stack 4 cubes so every side shows R, G, B and Y once
- 331,776 ways to stack them; only 1 tower works (seen 8 ways)
- Key fact: a cube is 3 pairs of opposite faces
  - one pair faces front/back, another faces left/right
  - so each cube makes one choice per axis
- Rule: across front+back, each colour appears exactly twice
  - same for left+right
- N cubes, N colours: NP-complete (Robertson & Munro, 1978)



our answer, checked with solves()

OUR ENCODING

Rules become an energy

- 24 qubits = 4 cubes × 2 axes × 3 pairs (one-hot blocks)
  - block 010 means "pair 1 faces this axis"
- Hamiltonian (Z and ZZ terms) adds a penalty for:
  - a colour not appearing exactly twice on an axis
  - one pair used for both axes (μ · x · y)
- Energy 0 only for a solved tower
  - lowest energy = the answer: QAOA looks for it

16.8 M

bitstrings on 24 qubits

6,561

valid choices (one 1 per block)

2 of them are correct

OUR ALGORITHM

Constrained QAOA: never leave the valid choices

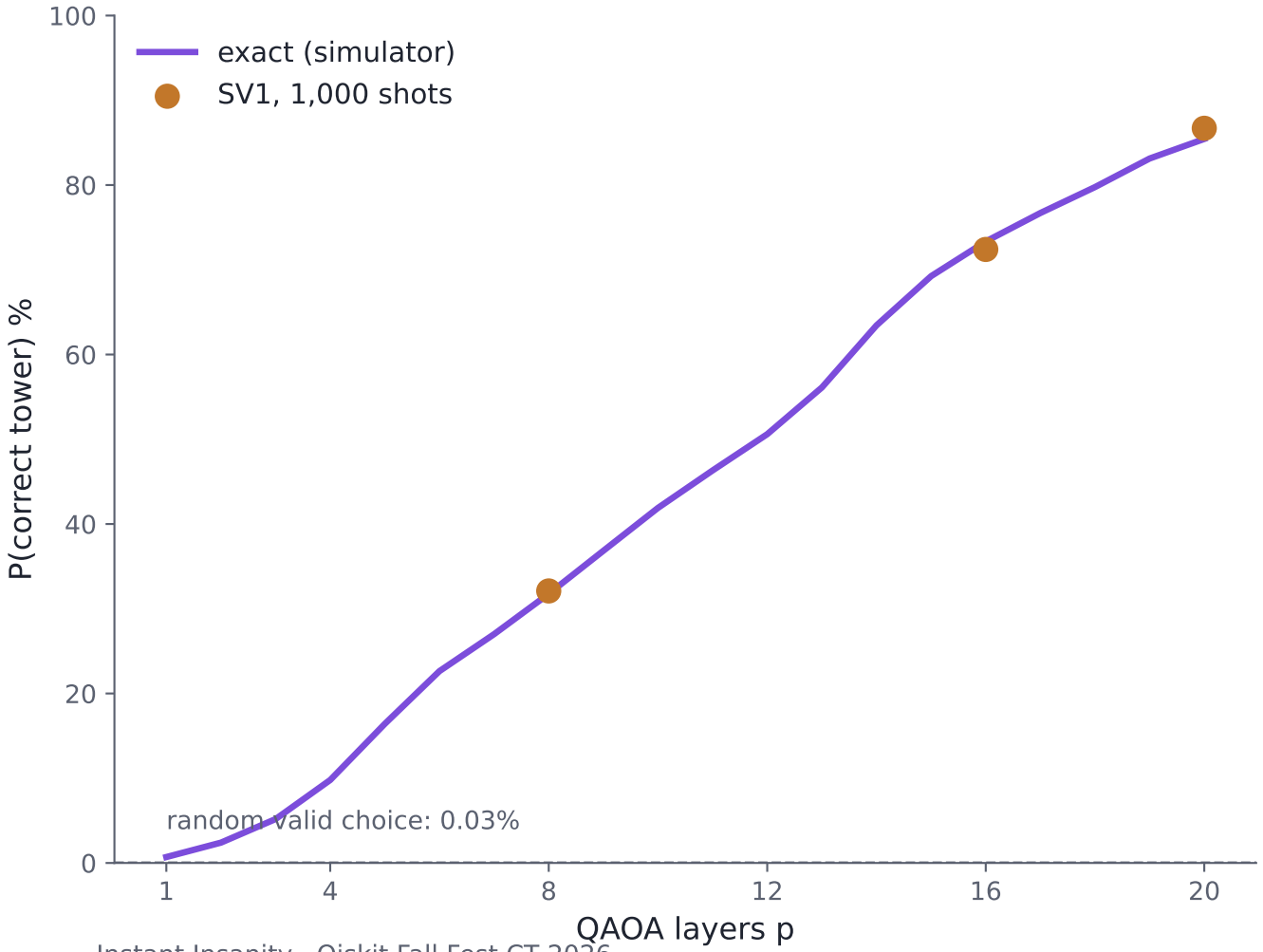
- Start: W state on every block = equal mix of its 3 options
- Repeat p times:
  - cost step: phase each choice by its energy (rz, rzz)
  - XY mixer: move the 1 inside a block, never add or remove one
- Circuit only ever explores the 6,561 valid choices
- Tuning: our own exact simulator of those 6,561 states
  - matches Qiskit Aer on the real 24-qubit circuit to 1e-17
  - COBYLA tunes the angles, adding one layer at a time
- Then: run the tuned circuit on AWS SV1 through qBraid

How we know it is right

- Hamiltonian = classical cost on all 6,561 valid choices (gap 0)
- Energy-0 choices are exactly the 2 the tower checker accepts
  - everything else costs ≥ 2: no wrong answer ties
- Fast simulator = Aer on the real 24-qubit circuit (≤ 1e-17)
  - and the real circuit puts no probability outside valid choices
- Decoder tested on the known answer (Qiskit's reversed bit order)
- Every shot decoded and checked with solves() before it counts
- Same checks for the 11-qubit encoding and for N = 5, 6

RESULTS ON SV1

More layers, more correct towers



p	SV1	exact	CX	depth
8	32.1%	31.8%	2,400	871
16	72.4%	73.3%	4,736	1,719
20	**86.7%**	**85.4%**	**5,904**	**2,143**

24 qubits · 1,000 shots per run

p = 20: about 2,800× a random valid choice

Gains level off after p ≈ 20 (89% at p = 25)

Job IDs for every run: README

EXPERIMENTS

Scaling: fewer qubits, bigger puzzles

11 qubits instead of 24 (3 bits per cube)

	one-hot	compact
qubits	24	11
mixer	XY (constrained)	rx + penalty
largest term	2 Z's	6 Z's
CX per layer	~290	~480
P(correct), p = 20	85.4%	33.8%
1,000 shots, p = 20	86.7% (SV1)	30.2% (Aer)

Fewer qubits cost success rate: the plain mixer is much weaker than XY.

Bonus: N cubes, N colours (p = 6, exact)

N	qubits	P(correct)	random valid
4	24	22.6%	0.03%
5	30	6.1%	0.0034%
6	36	2.9%	0.0004%

SV1 caps at 34 qubits:

one-hot fits $N \leq 5$, compact fits $N \approx 11$

TAKEAWAYS

What we learned, and what's next

- Building the rules into the circuit is the biggest lever
  - 86.7% correct on SV1 vs 0.03% for a random valid choice
- Not a speedup: brute force solves 4 cubes in 0.1 s
  - our tuning itself simulates the 6,561 choices classically
- Depth is free on SV1, not on hardware
  - ~5,900 CX at p = 20 would drown in noise on today's QPUs
- Next: a constrained mixer for the 3-qubit-per-cube code
  - fewer qubits and no leakage: larger N on SV1
- AI use: Claude Code wrote Sections 7–8 and experiments/; every step checked classically